

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (E & C) Examination

Nov/Dec 2013

EC – 312 Digital Signal Processing

Date: 07.12.13, Saturday

Time: 01:30 p.m. to 04:30 p.m.

Maximum Marks: 70

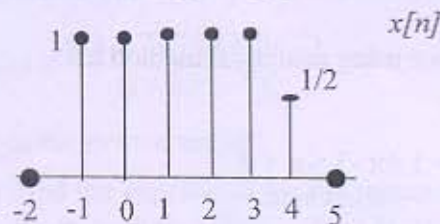
Instructions:

- (1) All questions are compulsory.
- (2) Indicate clearly, the option you attempt along with its respective question number.
- (3) Figures to right indicate marks
- (4) Rough work is done in the last page of main answer book; don't write anything on the question paper.

SECTION-I

Q-1 A A discrete time signal $x[n]$ is shown in figure below:

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Skeeth and label carefully each of the following signals:

- (i) $x[n-2]$
- (ii) $x[4-n]$
- (iii) $x[2n]$
- (iv) $x[n]u[2n-1]$
- (v) $x[n-1]\delta[n-3]$

B Answer the following questions:

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- (i) What is BIBO (Bounded input and bounded output) stability? Explain the condition for stability of LTI sytem.
- (ii) Why linear convolution is important in digital signal processing?
- (iii) Write the advantage of digital signal processing over analog signal processing.

Q-2 Attempt any two.

A For following two systems determine whether the systems are (i) satable, (ii) 6 Causal (iii) Linear (iv) tim einvariant and (v) memoryless.

$$(a) T(x[n]) = \sum_{k=n_0}^n x[k] \quad (b) T(x[n]) = e^{x[n]}$$

B A discrete time system is given by the expression

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$$x[n] = n \left(-\frac{1}{2} \right)^n u[n] \otimes \left(\frac{1}{4} \right)^n u[-n],$$

Find the z-transform.

C Let an analog signal $x_a(t) = 10 \cos 200\pi t$. If the sampling frequency is 150 Hz

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find the discrete time signal $x(n)$. Also find the alias frequency corresponding to $F_s = 150$ Hz.

Q-3 Attempt any two.

A What is sampling theorem? How to reduce aliasing?

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B Compute convolution of following sequence using analytical method of convolution.

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$$x(n) = n/3 \text{ for } 0 < n < 6 \\ 0 \text{ elsewhere.}$$

$$h(n) = 1 \text{ for } -3 < n < 3 \\ 0 \text{ elsewhere.}$$

C Prove any three properties of Z-transform.

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SECTION-II

Q-4

- A Compare the fixed point arithmetic and floating point arithmetic. 5
 B When input to the LTI system is $x[n] = 5u[n]$, the output is 6

$$y[n] = -2\left(\frac{1}{2}\right)^n u[n] + 3\left(-\frac{3}{4}\right)^n u[n]$$

- (i) Find the system function $H(z)$. Plot the poles and zeros of $H(z)$ and indicate the region of convergence.
 (ii) Find the impulse response of the system for all values of n .

Q-5

Attempt any two.

- A A discrete time LTI system has the system function 6

$$H(z) = \frac{(1 + 0.2z^{-1})(1 - 9z^{-2})}{(1 + 0.81z^{-2})}$$

- (i) Is the system stable?
 (ii) Find the expression for the minimum phase system $H_1(z)$ and all pass system $H_{ap}(z)$ such that $H(z) = H_1(z) H_{ap}(z)$
 B Obtain the direct form II and cascade realization of the system 6
 $Y[n] = -0.1y[n-1] + 0.2y[n-2] + 3x[n] + 3.6x[n-1] + 0.6x[n-2]$.
 C Determine $H(z)$ using impules invariant method at 5 Hz sampling frequency from $H_a(s)$ given below:

$$H_a(s) = \frac{2}{(s+1)(s+2)}$$

Q-6

Attempt any two.

- A Find the direct form I and direct form II realization of a system represented by 6
 $y(n) - 3/4 y(n-1) + 1/8 y(n-2) = x(n) + 1/2 x(n-1)$
 B A digital low pass IIR filter is to be design with buttorworth approximation using 6
 bilienear transformation technique. Find the order of the filter to meet the following specification. Assume $T = 1$.
 (i) passband magnitude is constant with in 1 dB for frequency below 0.2π .
 (ii) stopband attenuation is greater than 15 dB for the frequency between 0.3π to π .
 C Compute the 4 point DFT of following window function 6
 $w(n) = u(n) - u(n-N)$